

Single-Pass Drone Video to a Complete 3D Model on One GPU:

Overlap-Band Keyframes, a Parallax Test for “Is It 3D?”, Structure from Optical Flow and Fused Depth

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Abstract

We present an end-to-end system that turns a single drone pass — one video, no flight planning, no ground control — into a textured 3D mesh and a dense point cloud and, when a flight log is available, georeferenced metric outputs in the formats problem statement 26158 asks for (OBJ, PLY, LAS, GeoTIFF, glTF, FBX), with a web viewer. Every stage runs on one GPU. A keyframe selector formalises the intuition that consecutive views must be *similar enough to match and different enough to triangulate*: it chains optical flow to measure co-visibility and picks the sharpest frame inside an overlap band $[\tau, \tau + \delta]$. Before any reconstruction it answers whether a pass contains recoverable 3D structure at all, from rotation-compensated parallax measured with *direct* keyframe-pair flow and normalised by an estimated noise floor. The same flow then replaces feature matching: tracks from direct RAFT flow give camera poses within 0.19% of the flight extent and 0.28° of SIFT structure from motion, triangulated flow against wide keyframe baselines gives dense depth, a monocular prior calibrated to it fills what flow cannot see, and a GPU TSDF fuses the views. A 187s video is processed in 243s against a budget of 281s (15 minutes per 10-minute video), covering 0.75 of the non-sky pixels its keyframes saw. On a straight single-pass track the usual 7-DoF GPS alignment silently rolls the model about the flight axis; levelling on the ground plane and fitting four degrees of freedom removes the failure, and with synthetic GPS on real models the exported mesh is within 0.43–2.06 m of the truth near the track. A slower profile trains depth-supervised 3D Gaussian Splatting. All numbers are produced by scripts in the repository from the run directories.

1 Introduction

Operational mapping — disaster response, inspection, reconnaissance — often has one chance to fly over a site. Classical photogrammetry assumes the opposite: planned grids, high overlap, several passes and ground control points. Problem statement 26158 asks for an accurate, textured, georeferenced 3D model from a *single pass* of a moving UAV [1].

A single pass is hard for three distinct reasons, and

each one maps to a component of our system.

1. **Which frames?** A video carries thousands of highly redundant frames and a few bad ones (motion blur, compression, fades). Sampling at a fixed rate is the common default; it either wastes compute or, where the drone moves fast, leaves gaps that break the reconstruction into pieces (§3.3).
2. **Is it 3D at all?** A drone that yaws on the spot, or looks at a distant or planar scene, produces video in which a homography explains every frame pair. No amount of computation recovers depth from such footage; the system should say so before spending it (§3.4).
3. **Where is it?** A single pass is nearly a straight line. Aligning the model to GPS through camera centres alone leaves the roll about that line undetermined (§3.8).

A fourth constraint is time: the statement’s budget is 15 minutes for a 10-minute video, and it scores processing speed (20%) next to accuracy (30%) and completeness (20%). Around these we build two GPU pipelines. The *fast profile*, the deliverable, reuses the selector’s optical flow (RAFT [2]) for structure from motion (pycolmap’s global mapper [3]) and dense depth, completes it with Depth Anything V2 [4], and fuses a textured mesh and point cloud in a GPU TSDF [5, 6] (§3.10). The *accurate profile* uses spirula-studio’s SIFT structure from motion and 3D Gaussian Splatting [7, 8] with the Marigold v2 depth prior [9] aligned to, and scored against, the SfM points.

Contributions. (i) An overlap-band keyframe selector over chained dense flow with an explicit multiplicity model, and pass segmentation for edited footage. (ii) A per-pass 3D verdict from direct-flow residual parallax with a data-driven noise floor, validated on a synthetic pure-rotation control; we report where Torr’s GRIC [10] fails on flow correspondences. (iii) A measurement of chained versus direct optical-flow correspondence error against exact ground truth. (iv) A ground-levelled 4-DoF georeferencing rule with a significance test for sloped terrain, leave-one-out error and a jackknife scale uncertainty. (v) Structure from motion and dense depth from the selector’s optical flow instead of features, with

a measurement of the triangulation baselines a single pass offers, and a study of the TSDF truncation band as the lever on completeness. (vi) An engineering study of fitting the budget on one GPU: a single decode for analysis and keyframes, a motion-adaptive analysis rate, RAFT without host syncs under CUDA Graphs, GPU texture baking, and per-stage time and utilisation recorded against the budget.

2 Related work

Structure from motion and splatting. Incremental SfM as in COLMAP [11] remains the reference for camera recovery; spirula-studio re-implements an incremental mapper on the GPU and adds sequence-aware matching for video [7]. 3D Gaussian Splatting [8] represents a scene as anisotropic Gaussians optimised by differentiable rasterisation; gsplat [12] provides the rasteriser we use for fly-through rendering.

Keyframe selection. Visual SLAM systems insert keyframes by co-visibility and tracking quality, and ORB-SLAM chooses between homography and fundamental-matrix initialisation by model scores [13]; Torr’s GRIC formalises that choice [10]. We use dense flow with the forward-backward consistency test of Sundaram et al. [14].

Monocular depth. Marigold [15] repurposed a latent diffusion model for affine-invariant depth; Marigold v2 [9] fine-tunes the Qwen-Image-Edit diffusion transformer [16] to predict affine-invariant log depth in one step. Depth Anything V2 [4] is a strong discriminative alternative.

Compositional reconstruction. WorldSculpt [17] lifts per-object priors into cluttered scenes from posed, masked views; we discuss it as a downstream extension in §5.

3 Method

3.1 GPU ingest

Frames are decoded by NVDEC through ffmpeg (`-hwaccel cuda`), resized on the GPU (`scale_cuda`) and leave the decoder as raw NV12; conversion to RGB (BT.709, limited range) runs in PyTorch on the GPU. A background thread streams chunks so decoding overlaps optical flow. Keyframes are later re-decoded at full resolution, selected by timestamp in groups of 24 on two concurrent decoder sessions, cropped on the GPU and encoded with nvJPEG.

3.2 Pass segmentation

For analysis frames $I_0, \dots, I_{N-1}$ (long side 640px, ≈ 12 fps) we compute RAFT flow $\mathbf{f}_t$ from t to $t+1$ and $\mathbf{b}_t$ back, and the share c_t of pixels passing the forward-backward test $\|\mathbf{f} + \mathbf{b}\|^2 < 0.01(\|\mathbf{f}\|^2 + \|\mathbf{b}\|^2) + 0.5$ [14].

Inside a continuous camera move c_t stays high; a cut, fade or dissolve collapses it. A *pass* is a maximal run with $c_t \geq 0.35$, no dark or flat frame, and at least 2s; its ends are trimmed while they are darker than $0.8\times$ the pass median, removing fade-ins and fade-outs (a fade kills tracks, so without trimming it attracts keyframes, and a fade frame is a meaningless held-out view). Letter-box bars are found before analysis from the per-row and per-column maximum luma over twelve seek-sampled frames and cropped on the GPU: static black would otherwise count as perfectly tracked content, inflating overlap, and as trivially correct pixels, inflating held-out PSNR. Real footage is often edited; a true single pass yields one pass, and SfM rejoins passes that turn out to share views.

3.3 Overlap-band keyframes

A grid G of points (8px spacing) on keyframe K is chained through the consecutive flows; a track dies when it fails the consistency test or leaves the frame. With $A_t \subseteq G$ the tracks alive at frame t ,

$$\text{ov}(K, t) = \min\left(\frac{|A_t|}{|G|}, \frac{|\mathcal{C}(A_t)|}{|\mathcal{C}|}\right), \quad (1)$$

where $\mathcal{C}$ are the 32px cells of frame t and $\mathcal{C}(A_t)$ those containing a live track. The first term falls as K ’s content leaves the view (forward motion), the second as new content enters it (backward motion), so the measure is symmetric in effect. The next keyframe is the sharpest frame (Laplacian variance relative to a 2s rolling median) among those with $\tau \leq \text{ov}(K, t) \leq \tau + \delta$ before the curve first drops below τ ; defaults $\tau = 0.75$, $\delta = 0.1$. If the overlap drops below τ within one step, the next frame is taken to keep the chain connected.

Multiplicity. Under uniform motion a point stays in view for about $1/(1 - \tau)$ keyframe spacings, so each surface point is observed by $m \approx 1/(1 - \tau)$ keyframes ($m = 4$ at $\tau = 0.75$) — the “two or more frames with a proper threshold of similarity” that triangulation needs, stated as a design parameter. We report the measured multiplicity, $1 + 2\bar{L}$ with $\bar{L}$ the mean forward track length in keyframes.

Cost. Only consecutive pairs are flowed, so analysis is linear in video length. An overlap matrix $M_{i,d} = \text{ov}(i, i+d)$ for all frames and $d \leq d_{\max}$ is built in one sweep: a fresh grid starts at every frame and all live grids advance together in a ring of $d_{\max}$ slots, one batched `grid_sample` per frame and a single read-back; selection is then a CPU walk over M .

3.4 Is it 3D? A parallax test

For each keyframe i we take its widest partner $j \leq i+6$ with $\text{ov} \geq 0.3$, compute RAFT flow *directly* between I_{K_i} and I_{K_j} in both directions, and keep consistent grid correspondences $(\mathbf{x}_n, \mathbf{y}_n)$. On a batch of all pairs we fit, by normalised DLT [18] with Cauchy IRLS, a homography H and a rank-2 fundamental matrix F . The residual $\|\mathbf{y} - H\mathbf{x}\|$ is rotation-compensated parallax: zero for

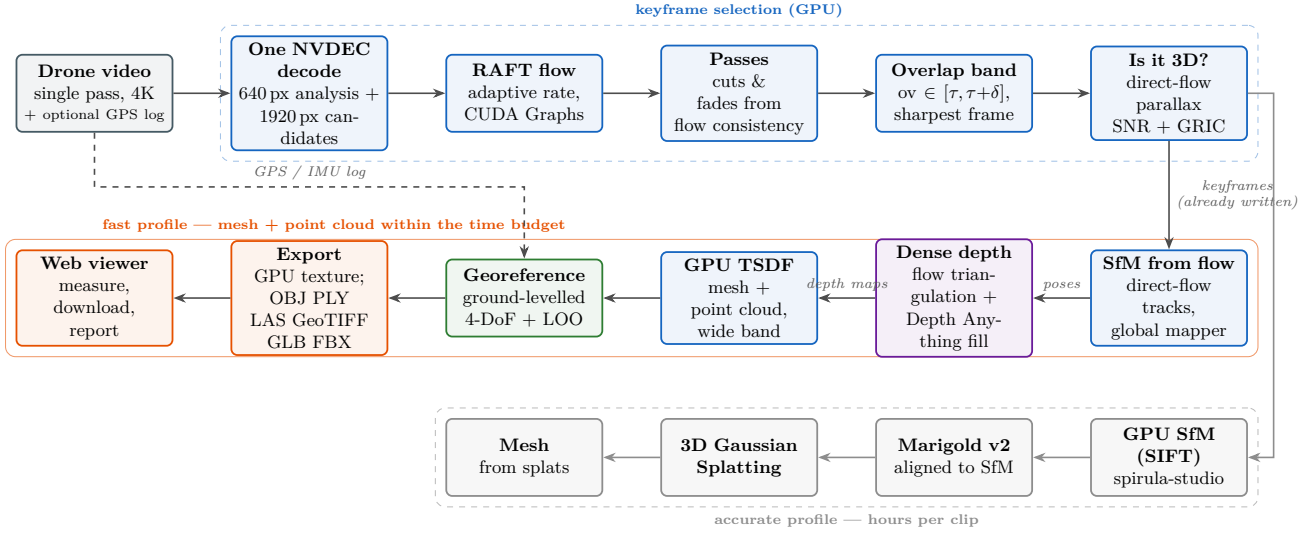


Figure 1: Pipeline. The keyframe selection is shared; the fast profile (the deliverable) reuses its optical flow for poses and depth and fuses a textured mesh and point cloud, the accurate profile trains Gaussian splats. Everything runs on one GPU; stages exchange files only, so any stage can be re-run alone.

pure rotation or a planar scene, growing with depth variation times baseline. We estimate the per-coordinate noise $\hat{\sigma} = 1.4826 \text{ median } \sqrt{e_F^2}$ from F 's Sampson residuals e_F — pure noise whenever F is correct, which includes degenerate motion — and normalise

$$\text{SNR} = \frac{\text{median}_n \|\mathbf{y}_n - H\mathbf{x}_n\|}{\sqrt{2}\sqrt{2\ln 2} \hat{\sigma}}, \quad (2)$$

where the denominator is the median transfer residual that isotropic noise in both views leaves, so $\text{SNR}=1$ means no measurable parallax. A pass is *3D* when its median SNR over keyframe pairs is at least 2 and GRIC [10] (both models scored on geometric error with $\hat{\sigma}$) prefers F on at least 60% of pairs; *degenerate* when the SNR is below 1.5 and its 90th percentile below 2.

Why direct flow. Chaining consecutive flows makes long-range correspondence cheap but its error accumulates with a systematic per-step bias (§4, Fig. 2); over a keyframe span the drift reaches the size of the parallax being measured and is correlated across points, which the extra freedom of F partly absorbs. Direct flow keeps the error near 0.1 px at the same gaps.

3.5 Structure from motion

Keyframes are written per pass (`images/pass_k/`) and reconstructed by spirula-studio's GPU SfM with each pass declared a sequence, so neighbours are matched and trusted first, plus GPU pair selection for loop closure. We use a *single-focal* radial camera per pass: drone cameras have square pixels, and with independent f_x, f_y the weak geometry of forward or orbital motion let the two drift 10–20% apart for no gain in reprojection error (§4). Separate passes of an edited video become separate models unless they share views.

3.6 Monocular depth prior

Marigold v2 [9] predicts, per keyframe, $p(\mathbf{u})$ with $\log d(\mathbf{u}) = ap(\mathbf{u}) + b$ for unknown a, b . It runs as upstream: 4-bit NF4 base transformer (what its LoRA was trained on), rank-128 LoRA and fine-tuned VAE decoder, one rectified-flow step, bf16 autocast, at 1024 px; the quantised base is cached so a load reads ≈ 11 GB rather than re-quantising 39 GB. For every registered image the SfM tie points give depths z_n at pixels $\mathbf{u}_n$.

Affine is not enough at landscape scale. A Cauchy-IRLS fit of $\log z_n = ap(\mathbf{u}_n) + b$ works well where the tie points span a narrow depth range, but on drone footage the prior *compresses the far field*: on Jal Mahal it places the far shore at the depth of the palace and the mountain ridge barely beyond it (Fig. 3), so a single slope cannot fit both. We therefore calibrate each image with a monotone map: medians of $\log z$ in quantile bins of p , made non-decreasing by weighted pool-adjacent-violators, linear with the affine slope outside the knots. It keeps the prediction's ordering and lets the mapping bend. Both calibrations are scored on *held-out* tie points (5-fold cross-validation per image).

Where the prior saturates. In a distant pass every tie point may sit where the prediction has saturated at its far end; the correlation between p and $\log z$ is then near zero or negative. The robust fit returns a non-positive slope and the stage writes no depth map for that image, so training is not supervised with wrong geometry.

The calibrated linear depth supervises splat training through spirula's Pearson-correlation depth loss (weight 0.05). Because that loss is affine-invariant in *linear* depth, the calibration — which only SfM can provide — shapes the supervision itself, not merely its reporting. *What* is supervised matters as much: outside the range of predictions the tie points cover (sky, featureless far

field) the calibration only extrapolates, and a depth invented for the sky asks the trainer to put splats there. The mask therefore keeps pixels whose prediction lies within the 0.5–99.5% range of the tie-point predictions (plus 5% margin) and below $3\times$ the 99th-percentile SfM depth. The canonical run below was trained before this mask existed (only the far cut, which removed nothing); the masked maps are evaluated as an ablation (§4).

3.7 Gaussian splatting and meshing

Each SfM model with at least 8 images trains a 3DGS model in spirula-studio (30k steps, half resolution, bilateral-grid exposure correction), with every 8th keyframe held out for evaluation; meshes are extracted from the trained splats with vertex colour and texture atlases.

3.8 Georeferencing a straight pass

With GPS positions $\mathbf{g}_n$ (local ENU) for camera centres $\mathbf{c}_n$, the usual fit is the 7-DoF similarity of Umeyama [19]. If the $\mathbf{c}_n$ are nearly collinear the rotation about the line is unobservable and GPS noise sets it; the in-sample RMSE still looks like GPS noise because the unobservable direction does not affect it. We instead estimate the model’s ground normal $\mathbf{n}$ by RANSAC on the sparse points (oriented towards the cameras), rotate $\mathbf{n}$ to $+z$, and fit yaw, scale and translation — scale and yaw by complex least squares on horizontal coordinates, the vertical offset as a mean residual. The full similarity is used only if the track spans two dimensions *and* its tilt against $\mathbf{n}$ exceeds 5° by three jackknife standard errors, i.e. GPS proves the ground is sloped. We report leave-one-out RMSE rather than in-sample RMSE, and a jackknife relative standard error of the scale as the scale check.

3.9 Keeping the GPU busy

Stock torchvision RAFT rebuilds its correlation lookup offsets on the CPU in each of its 12 refinement iterations, divides by a CPU tensor, builds coordinate grids on the CPU, and upsamples the flow after every iteration although inference keeps only the last; at 640px a call cost roughly the same whatever the batch size, a sign of launch- and sync-bound execution. Removing these (identical arithmetic, differences $\leq$ bf16 noise) and replaying the network from a CUDA Graph raised throughput —fold. The selector’s tracking avoids boolean-mask indexing, whose `nonzero` forces a host sync, in favour of scatter and redirected writes. Every stage runs under an NVML sampler recording utilisation, power, memory and the presence of other processes on the GPU, so no timing is reported without its context.

3.10 A fast profile for the time budget

The accurate profile spends hours per clip on splat training and meshing; the problem statement asks for a mesh or point cloud within $1.5\times$ the video length. The

fast profile keeps the selector and replaces everything after it.

One decode. The analysis decode also delivers every analysis frame at the output size: ffmpeg splits the NVDEC stream, scales both branches with `scale_cuda` and stacks them in one NV12 frame, so one pipe carries both and cannot deadlock; the planes are split before colour conversion so chroma does not bleed across the seam, and candidates are nvJPEG-encoded as they stream. The chosen keyframes are then files already written, where a second, seek-heavy 4K pass took 31 s for a 55 s clip. Twelve frame pairs one nominal analysis step apart, sampled across the video, give the median flow per step; the analysis stride grows until a step moves about 2% of the width (at least 3 analysis frames per second). The sample videos move 0.5–3 px per step at 12 fps.

Structure from flow. For keyframes $K_1, \dots, K_n$ of a pass at 960px we compute direct RAFT flow from each K_i to $K_{i+1}, \dots, K_{i+3}$ in both directions. Points seeded on a grid where no track is alive move from K_i to K_{i+1} by sampling the flow at their sub-pixel position (forward–backward consistent only); each step is also predicted by the direct flow from K_{i-1} and K_{i-2} , and a track whose chained position disagrees with a prediction by more than 1 px ends, while agreeing predictions are averaged, so chained drift (§3.4) cannot accumulate. The tracks are written as COLMAP keypoints at full resolution with matches between keyframes up to four apart, verified by RANSAC and mapped by the global mapper [3], capped at about twelve tracks per image. The incremental mapper’s defaults — a 16° initialisation angle, 1.5° to triangulate — reject whole single passes, whose nearby keyframes are $0.1\text{--}1^\circ$ apart at the scene.

Depth from flow, completed by a prior. With the poses fixed, every consistent flow vector from a reference keyframe to neighbours $d \in \{2, 4, 8, 12\}$ keyframes away is triangulated by the midpoint of the two rays; a depth is kept if the rays meet at $\geq 0.5^\circ$ and it reprojects within 1.5 px in both views. Consecutive keyframes are too close for this — their median triangulation angle on Jal Mahal is $0.04\text{--}0.4^\circ$, twelve apart $0.6\text{--}1^\circ$ — hence the wide gaps. Candidates from all neighbours are fused per pixel (median, then an angle²-weighted mean of those within 5%, at least two views). What flow cannot triangulate (water, far field, low texture) is filled by Depth Anything V2 [4], calibrated per image to that image’s triangulated pixels with the monotone map of §4; its sky is exactly zero disparity, and those pixels are never filled.

Fusion, texture and export. All depth maps are integrated in an Open3D GPU TSDF [5, 6] whose voxel is three pixel footprints at the median depth and whose truncation band is twelve voxels (§4.7). The mesh is textured from the 1920px keyframes on the GPU without UV unwrapping: triangles are packed two per square cell of a 4096^2 atlas, each triangle takes the

keyframe that sees it most frontally and closely without occlusion (a z-buffer of triangle centroids per view), and each texel samples that keyframe at its point on the triangle. Exports are PLY (full density), GLB, textured OBJ/GLB and FBX (a copy with at most 600k triangles), LAS, and a DSM and orthophoto rasterised from the point cloud; with georeferencing, LAS and GeoTIFF are in UTM with the EPSG code (ENU $\rightarrow$ ECEF $\rightarrow$ UTM, not ENU offsets). A self-contained three.js viewer opens at the middle keyframe’s viewpoint and measures distances and height differences.

Time. Every stage’s wall time is recorded next to the budget. Two host-side traps cost more than any kernel: the login shell’s `OMP_PROC_BIND=spread` made the first OpenMP runtime pin each process, and every thread and child it later started (the mapper, `ffmpeg`), to core 0 — the global mapper got 0.13 cores until re-pinned — and torch’s CPU path spent 48s calibrating 97 depth maps with 24 threads per small operation, 3s in numpy.

4 Experiments

Setup. One NVIDIA A100 80 GB PCIe, shared with other users’ jobs throughout (each stage’s record states the GPU’s occupancy); PyTorch 2.14 with CUDA 13.2, Python 3.14, `ffmpeg` 9.0 (NVDEC/NVENC), `spirula-studio` 2026.9.24 (Vulkan backend). Data: fifteen public drone videos (Table 1); they carry no flight logs.

Table 1: Sample footage (public YouTube re-encodes, VP9).

Video	Resolution	fps	s
ANGKOR WAT Breathtaking Cinemati	3840×2160	24.00	193
Above clouds	2560×1440	50.00	115
Aerial Views of Rural Riches	2560×1440	23.98	19
Breathtaking Aerial Views of Hanoi	3840×2160	29.97	103
Colosseum like never before	3840×2160	29.97	135
Eiffel Tower Drone 4k	3840×2160	29.97	306
FPV Drone Flight through Beautiful GoPro	3840×2160	30.00	67
GoPro	1080×1920	29.97	18
HAGIA SOPHIA 4K	3840×2160	29.97	85
Jal Mahal, Jaipur, India, Drone Ci	3840×2160	24.00	55
Notre Dame Drone Paris 4k	3840×2160	29.97	214
Petronas Twin Tower in FPV drone	3840×2160	59.94	35
Qutub Minar, New Delhi, India 4K D	3840×2160	29.97	187
Reichstag Berlin in 4K	3840×2160	30.00	130
The Messiah	3840×2160	29.97	94

4.1 Passes, keyframes and the 3D verdict

On Jal Mahal the flow signal splits the edit into 5 passes at its cuts and fades, and the band selects 213 keyframes; the mean overlap with the previous keyframe lies in 0.74–0.86, inside or just above the target band, and every pass is judged *3d* (Table 2). Decoding and flow for the 658 analysis frames took 37s, selection 12s and 4K extraction 45s.

Negative control. A synthetic clip of a camera rotating about its centre over one real 4K frame

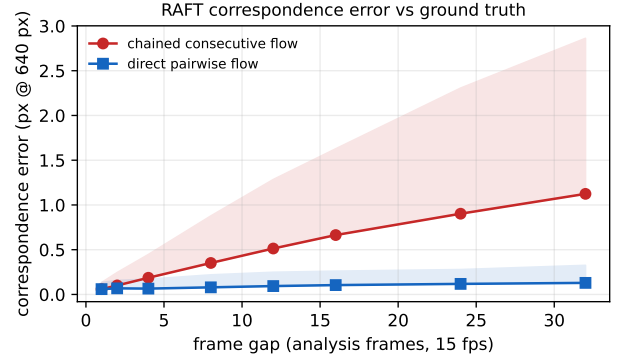


Figure 2: Correspondence error against exact ground truth on the rotation control. Chained consecutive flow drifts almost linearly (1.12 px median at 32 frames); direct pairwise flow stays at 0.13 px.

(`tools/make_degenerate_clip.py`) has no parallax by construction. It yields SNR 1.24 (median residual 0.02°) and the verdict *degenerate*; GRIC nonetheless preferred *F* on 60% of its pairs, which is why GRIC only confirms the SNR decision.

4.2 Structure from motion

With the same SfM settings, 1 fps sampling registers 40 of 55 frames in 2 disconnected models (the largest 22), and uniform sampling at the selector’s budget of 241 frames registers 237 but splits into 4 models (largest 98). The overlap band registers all 241 (largest 155); after cropping the letterbox and trimming fades the canonical run registers 213 of 213 (Table 4). A model per group of passes is expected: the passes of a cinematic edit are cut from different flights and need not overlap. One focal length per pass keeps the reprojection error while OpenCV’s independent f_x, f_y drift apart (Table 5).

4.3 Monocular depth prior

Table 6 scores each calibration on tie points it was not fitted to. The monotone map lowers held-out AbsRel on every model with depth; the gain is largest where the scene spans a wide depth range (Fig. 3), which a single log-affine map cannot follow. Model 3 (30 keyframes, the far lake-side pass) has no image whose prediction correlates positively with its tie-point depths; it is trained without depth rather than with a wrong one.

4.4 Gaussian splatting

4.5 Georeferencing a straight pass

None of the sample videos carries a flight log, so georeferencing is evaluated in simulation (`experiments/georef_study.py`): a 300m track at 80m altitude over flat ground with buildings, the SfM model hidden behind a random similarity, horizontal GPS noise, and a sideways deviation of the track from a straight line of 0–60m; the error is measured at ground points 100m off the track. (Sloped

Table 2: Pass segmentation and keyframes per run. Overlap: mean co-visibility with the previous keyframe; views: measured views per point; SNR: median direct-flow parallax SNR (1 = none).

Run	Pass	Time (s)	KF	Overlap	Views	SNR	Verdict
jal_mahal	0	0.0–7.4	30	0.77	6.9	22.4	3d
jal_mahal	1	8.0–14.1	14	0.79	5.2	3.6	3d
jal_mahal	2	14.2–23.8	59	0.77	7.7	6.0	3d
jal_mahal	3	23.8–36.0	97	0.74	7.4	16.3	3d
jal_mahal	4	37.2–54.0	13	0.86	8.2	3.2	3d

Table 3: Structure from motion on the selected keyframes.

Run	KF	Registered	Models	Reproj. (px)	Time (s)
jal_mahal	213	213	4	0.79	255

Table 4: Frame sampling on Jal Mahal (same SfM settings).

Sampling	Frames	Registered	Models	Largest
1 fps (uniform)	55	40	2	22 no letterbox crop
uniform, same budget	241	237	4	98 no letterbox crop
overlap band (dev)	241	241	4	155 no letterbox crop
overlap band (canonical)	213	213	4	156 cropped, fades trimmed

ground, where the rule falls back to the similarity, is covered by the unit tests.) The levelled fit is never worse than the similarity (worst 2.1 m) and its metric scale error stays below 0.6 % (Table 8, Fig. 4).

4.6 Throughput and GPU use

4.7 Fast profile: mesh and point cloud within the time budget

Structure from flow. On the 97-keyframe orbit of Jal Mahal, tracks from direct RAFT flow register every keyframe; with 960 px tracks the global mapper’s camera centres agree with spirula’s SIFT reconstruction to 0.19 % of the flight extent and its rotations to 0.28° (median), with no feature extraction or matching (Table 10). Tracking at 640 px is cheaper but 2.5–3× less accurate. COLMAP’s incremental mapper needs its initialisation angle lowered from 16° to 2° to start on single-pass footage at all and is eight times slower here.

Fusion. The depth maps (triangulated, then filled by the calibrated prior) cover every non-sky pixel of the references, yet a TSDF with the usual 4-voxel truncation band keeps only part of the view: neighbouring views disagree slightly and their signed distances cancel. Widening the band recovers most of it at a small cost in depth error against the triangulated geometry (Table 11); the profile uses 12 voxels.

Table 5: Camera model on Jal Mahal: independent f_x, f_y (OpenCV) against one focal (radial).

Camera	Focal(s) per pass (px)	Reproj. (px)	Points
OpenCV (fx, fy free)	3112/2812, 2055/2080, 2827/3382, 2986/2754	0.761	96,940
radial (one focal)	3118, 2684, 2808, 3109	0.771	96,598

Table 6: Marigold v2 against SfM tie points, scored on held-out tie points (5-fold cross-validation per image, median over images): log-affine vs monotone calibration.

Run	Model	Aligned	AbsRel aff.	AbsRel mono.	δ_1 aff.	δ_1 mono.
jal_mahal	0	156/156	0.050	0.027	0.857	0.886
jal_mahal	1	14/14	0.009	0.006	1.000	1.000
jal_mahal	2	13/13	0.075	0.033	0.945	0.989
jal_mahal	3	0/30	—	—	—	—

Time and completeness. The budget is 1.5× the video length (15 minutes for a 10-minute video). Completeness is the share of each registered keyframe’s non-sky pixels whose ray hits the mesh. The motion-adaptive analysis rate costs a few points of completeness and brings the 187 s Qutub Minar video within budget (243 s of 281 s); the 55 s Jal Mahal edit, five camera moves over a lake, stays over it (147 s of 82 s). Measuring overlap against the points that survive the first tracking step (so that water does not force a keyframe every step) was worse on both videos (Table 12).

Georeferencing, end to end. Each Jal Mahal model receives a known model-to-world similarity at the real site and its keyframes GPS with 1.5 m horizontal and 3.0 m vertical noise; the pipeline’s georef and export stages then run unchanged. The levelled fit recovers rotation to a few tenths of a degree and scale to half a percent; the exported mesh is within 0.43–2.06 m (median) of the truth within 100 m of the track, and the error grows with distance, as yaw and scale error do (Table 13). LAS and GeoTIFF outputs carry EPSG:32643 (UTM 43N).

Table 7: 3DGS on held-out keyframes (every 8th keyframe never used in training).

Run	Model	Splats	PSNR	SSIM	LPIPS	Train (s)
jal_mahal	model_0, depth prior (v1 mask)	3,000,000	35.00	0.977	—	3822

Table 8: Georeferencing study (medians over 20 trials): off-track error of the similarity and of our rule, in-sample RMSE of the similarity, held-out RMSE of ours, metric scale error.

GPS σ (m)	Lateral (m)	Sim. (m)	Ours (m)	Sim. in-sample	Ours LOO	Scale err.
0.5	0	149.9	0.21	0.71	0.73	0.1%
0.5	5	5.5	0.19	0.69	0.71	0.0%
0.5	20	2.1	0.19	0.68	0.70	0.0%
0.5	60	0.7	0.17	0.72	0.74	0.1%
1.5	0	180.0	0.62	2.08	2.16	0.1%
1.5	5	19.5	0.54	2.10	2.18	0.1%
1.5	20	5.0	0.58	2.16	2.24	0.2%
1.5	60	1.9	0.63	2.08	2.16	0.2%
3.0	0	220.7	0.92	4.15	4.29	0.2%
3.0	5	27.0	1.11	4.00	4.15	0.2%
3.0	20	7.6	1.29	4.28	4.44	0.3%
3.0	60	2.8	1.14	4.29	4.44	0.2%
5.0	0	146.9	1.77	7.08	7.32	0.3%
5.0	5	24.4	2.13	7.13	7.37	0.5%
5.0	20	11.0	1.59	6.92	7.14	0.4%
5.0	60	5.4	2.08	6.83	7.07	0.6%

Table 9: Per-stage wall time and GPU use (NVML, 0.5 s samples). Own/host: peak GPU memory and host RSS of our process tree.

Run	Stage	s	Util.%	W	Own GPU (GB)	Host (GB)	Shared
jal_mahal	keyframes	109	83	256	10.6	6.3	yes
jal_mahal	sfm	256	83	213	9.9	4.2	yes
jal_mahal	depth	522	84	265	24.3	11.2	yes

Table 10: SfM from flow tracks on Jal Mahal pass 3 against SIFT SfM of the same keyframes (similarity-aligned).

Tracks	Mapper	Flow (s)	Map (s)	Reg.	Reproj. (px)	Centre (% ext.)	Rot. med. (°)	Rot. max (°)
640 px, 8 iterations	global	8.8	9.8	97/97	0.32	0.53	0.79	1.44
640 px, 12 iterations	global	13.1	18.7	97/97	0.32	0.55	0.66	1.35
960 px, 12 iterations	global	28.9	16.2	97/97	0.32	0.19	0.28	0.41
960 px, 12 iterations	incremental	28.9	129.1	97/97	0.35	0.40	0.42	1.59

Reference: spirula-studio SIFT SfM of all 213 Jal Mahal keyframes, 255 s for extraction, matching and mapping.

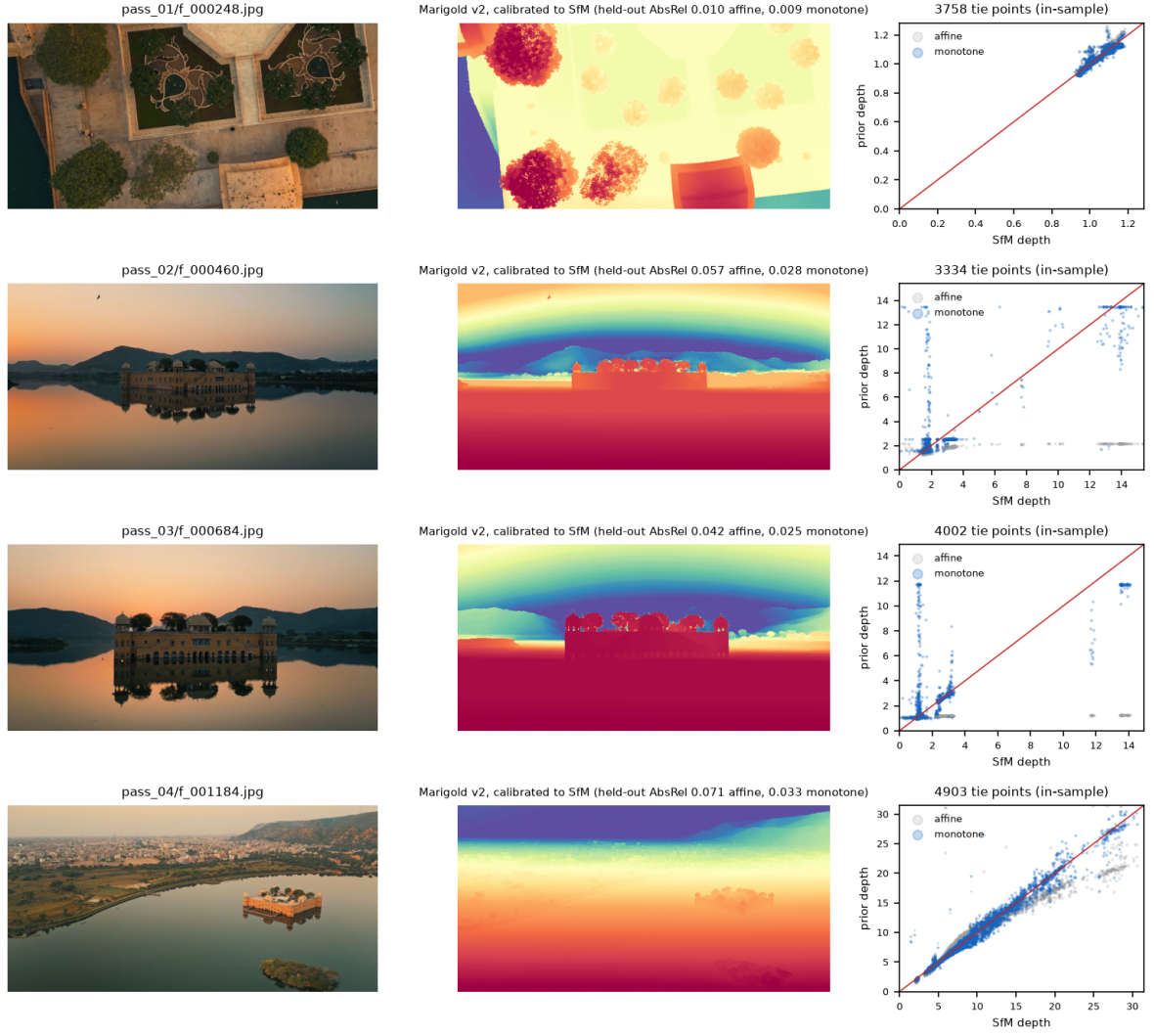


Figure 3: Marigold v2 on Jal Mahal keyframes (canonical run). Right: prior depth against SfM tie-point depth for the log-affine fit (grey) and the monotone calibration (blue). On the lake-facing passes the affine fit flattens the far shore and the ridge; the monotone map follows them (it also sends a few near points far). Held-out AbsRel for both is given above each depth map.

Table 11: TSDF truncation band on Jal Mahal’s two largest models (voxel = 3 pixel footprints at the median depth): view completeness, median depth error against the triangulated depth, triangles.

Model	Band (vox.)	Completeness	Depth err. (%)	Triangles
0	4	0.468	0.42	41,072
0	8	0.722	0.51	184,092
0	12	0.800	0.60	402,964
0	16	0.853	0.68	666,407
1	4	0.750	0.92	178,556
1	8	0.874	1.05	416,504
1	12	0.912	1.09	604,939
1	16	0.928	1.10	743,658

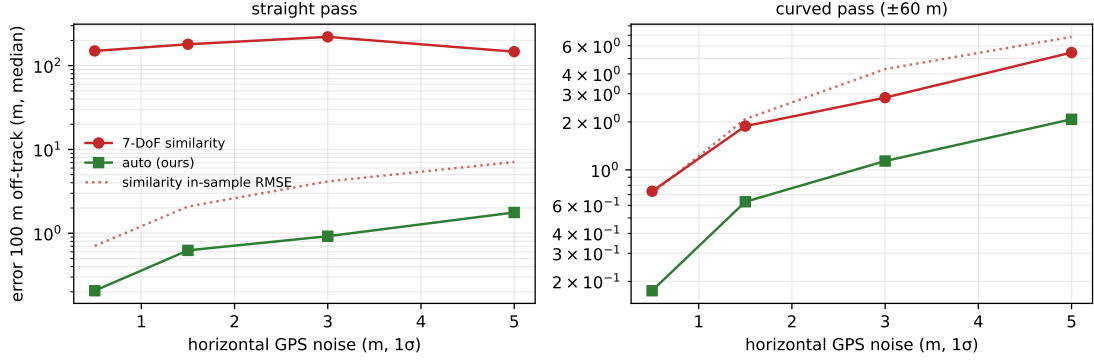


Figure 4: Error of ground 100 m off-track after GPS alignment (median of 20 synthetic scenes per point). The 7-DoF similarity is off by up to 221 m on a straight pass while its in-sample RMSE (dotted) only reflects GPS noise; the ground-levelled fit stays near the GPS noise.

Table 12: Fast profile end to end on a shared A100: keyframes, registration, view completeness, processing time and budget ($1.5\times$ video length). The adopted setting is the adaptive rate with absolute overlap.

Video	Analysis	Overlap	KF	Reg.	Compl.	Time (s)	Budget (s)
Jal Mahal	12 fps	absolute	213	213	0.82	262	82
Jal Mahal	adaptive, 3.0 fps	absolute	128	128	0.78	147	82
Jal Mahal	adaptive, 6.0 fps	relative	82	82	0.71	145	82
Qutub Minar	15 fps	absolute	110	102	0.81	307	281
Qutub Minar	adaptive, 3.3 fps	absolute	113	106	0.75	243	281 ✓
Qutub Minar	adaptive, 7.5 fps	relative	106	98	0.70	292	281

Table 13: Georeferencing with synthetic GPS on the real models: rotation error, scale error, held-out horizontal RMSE of the cameras, and median mesh error by distance from the flight track.

Model	Fit	Rot. (°)	Scale (%)	LOO (m)	<100 m	100–300 m	>300 m
0	yaw-scale	0.01	0.35	1.93	2.06	2.46	7.3
1	yaw-scale	0.23	0.07	1.99	0.86	1.19	8.7
2	yaw-scale	0.36	0.44	1.86	—	2.53	68.8
3	yaw-scale	0.02	0.47	2.11	0.43	0.72	—
4	yaw-scale	0.34	0.50	2.74	—	—	17.8
5	yaw-scale	0.06	0.50	1.97	—	—	—

5 Limitations

Budget. Timings were taken on an A100 shared with another user’s job throughout. The fast profile is within budget on the 187 s Qutub Minar video and $1.8\times$ over on the 55 s Jal Mahal edit (five camera moves over a lake, which defeats optical flow and forces dense keyframes). **Completeness.** About three quarters of what the keyframes saw is reconstructed; surfaces the single pass never faced cannot be, and the far field rests on a monocular prior. **No flight logs in the public samples.** The fifteen sample videos are YouTube re-encodes without GPS; georeferencing is evaluated in simulation and with synthetic GPS on real models, and runs on real data only when a DJI SRT, CSV, GPX or JSON log is supplied. Accuracy degrades with distance from the track (7–18 m at 300 m and beyond). **Edited footage.** Each pass of an edit becomes its own model. **Water and sky.** Reflections are reconstructed as mirrored geometry below the surface. **Dynamic objects** are not masked. **Licences.** Depth Anything V2 Large, the fast profile’s default prior, is CC-BY-NC-4.0; the Apache-2.0 Small model can replace it. **Compositional objects.** WorldSculpt [17] could turn the scene into per-object meshes given instance masks and boxes; we have not integrated it.

6 Conclusion

A single drone pass can be turned into a textured, georeferenced mesh and point cloud on one GPU within the problem statement’s time budget, provided the system chooses its frames by what they see rather than by the

clock, checks that the motion contains parallax before trusting it, reuses the optical flow it already computed for poses and depth instead of matching features, and aligns to GPS with the constraint a straight line cannot provide. The code, configurations and the scripts that regenerate every number in this paper are in the repository.

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